

Supplementary Information for “Closed-Form Absorbed-Power Dosimetry on the Human Body, 1 to 100 GHz”

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Abstract—This document presents the long-form derivations supporting the main paper. Section S1 derives the polarization-aware Fresnel surface law and the three conditions under which the polarization correction vanishes. Section S2 extends Azzam’s lossless high-index criterion to lossy biological tissue and quantifies the resulting angular variation across the IT’IS Cole–Cole catalog. Section S3 treats the layered tissue model behind the cube SAR below 6 GHz: Chew recursion of the generalized Fresnel reflection coefficient, the standing-wave SAR with forward, backward, and interference terms, layer-by-layer integration over the 10 g cube, the subsurface peak criterion, and the regimes where the layered model itself breaks down. Section S4 reports the per-frequency Mie residual analysis on body-scale lossy spheres with frequency-dependent IT’IS skin properties. Section S5 gives the anthropometric scaling of the whole-body compliance threshold. Numbering in this document is prefixed with S. Equations and figures cited without a prefix refer to the main paper.

S1. POLARIZATION-AWARE FRESNEL LAW

A. Setup

A monochromatic plane wave with time-averaged Poynting vector $\mathbf{S}_{\text{inc}} = \text{IPD } \hat{\mathbf{k}}$ illuminates a body whose surface Σ is locally flat on the wavelength scale. At a surface point $\mathbf{r}$ with outward normal $\hat{\mathbf{n}}(\mathbf{r})$, the local TE and TM unit vectors are

$$\hat{\mathbf{e}}_s(\mathbf{r}) = (\hat{\mathbf{k}} \times \hat{\mathbf{n}})/|\hat{\mathbf{k}} \times \hat{\mathbf{n}}|, \quad (\text{S1})$$

$$\hat{\mathbf{e}}_p(\mathbf{r}) = \hat{\mathbf{e}}_s \times \hat{\mathbf{k}}. \quad (\text{S2})$$

Decompose the incident electric field as

$$\mathbf{E}_0 = a_s \hat{\mathbf{e}}_s(\mathbf{r}) + a_p \hat{\mathbf{e}}_p(\mathbf{r}), \quad (\text{S3})$$

with complex amplitudes a_s, a_p that depend on the surface point because the local TE and TM bases rotate with $\hat{\mathbf{n}}(\mathbf{r})$ across the body. The local TE and TM intensity fractions are

$$|e_s(\mathbf{r})|^2 = \frac{|a_s(\mathbf{r})|^2}{|\mathbf{E}_0|^2}, \quad |e_p(\mathbf{r})|^2 = \frac{|a_p(\mathbf{r})|^2}{|\mathbf{E}_0|^2}, \quad (\text{S4})$$

and they satisfy $|e_s|^2 + |e_p|^2 = 1$ pointwise.

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B. Effective transmission

The Fresnel power-transmission coefficients on a planar interface between vacuum ($n_1 = 1$) and a lossy medium of complex refractive index $\tilde{n}$ are

$$T_s(\theta) = 1 - |r_s(\theta)|^2, \quad r_s = \frac{\mu - \xi}{\mu + \xi}, \quad (\text{S5})$$

$$T_p(\theta) = 1 - |r_p(\theta)|^2, \quad r_p = \frac{\tilde{n}^2 \mu - \xi}{\tilde{n}^2 \mu + \xi}, \quad (\text{S6})$$

with $\mu = \cos \theta_i$ and $\xi = \sqrt{\tilde{n}^2 - 1 + \mu^2}$, $\text{Re } \xi > 0$. For an incident field of (S3), energy conservation at the interface gives the effective transmission

$$T_{\text{eff}}(\mathbf{r}) = |e_s(\mathbf{r})|^2 T_s(\theta(\mathbf{r})) + |e_p(\mathbf{r})|^2 T_p(\theta(\mathbf{r})), \quad (\text{S7})$$

and the Absorbed Power Density (APD) at a visible point is $\text{APD}(\mathbf{r}) = \text{IPD } T_{\text{eff}}(\mathbf{r}) [\mu(\mathbf{r})]_+$.

C. Polarization splitting

Write

$$T_{\text{eff}}(\mathbf{r}) = T_{\text{avg}}(\theta) + \frac{1}{2} q(\mathbf{r}) \Delta T(\theta), \quad (\text{S8})$$

with $T_{\text{avg}} = (T_s + T_p)/2$ the unpolarized baseline, $\Delta T = T_p - T_s$ the polarization splitting, and $q(\mathbf{r}) = |e_p|^2 - |e_s|^2 \in [-1, 1]$ the local TM excess. Three conditions cancel the polarization term.

a) Circular polarization.: For circularly polarized illumination the field’s TE and TM intensities are equal at every point on every body, $|e_s|^2 = |e_p|^2 = 1/2$ pointwise, and $q \equiv 0$.

b) Random ensemble of linear polarizations.: For monochromatic linearly polarized illumination with random ensemble orientation, parameterize the polarization by an angle $\phi \in [0, 2\pi)$ with uniform distribution. The TE and TM projections at a point $\mathbf{r}$ scale as $\cos^2 \phi'$ and $\sin^2 \phi'$ with ϕ' a body-shifted angle, and $\langle \cos^2 \phi' - \sin^2 \phi' \rangle_\phi = 0$. The ensemble-averaged $\langle q \rangle$ vanishes.

c) Multipath averaging.: For a body in a multipath environment with N independent path orientations $\{\hat{\mathbf{k}}_n\}_{n=1}^N$ and equal power weights, write the body polarization directivity [1] as

$$D_B(\hat{\mathbf{k}}) = \frac{|\tilde{B}(\hat{\mathbf{k}})|}{2|\tilde{A}_\perp(\hat{\mathbf{k}})|}, \quad (\text{S9})$$

where $\tilde{B}$ is the off-diagonal Stokes component and $\tilde{A}_\perp$ the diagonal projected-area component, both at the body-integrated level. The variance of the polarization correction to the direction-averaged absorbed power is bounded by $D_B/\sqrt{2N}$. With $D_B = 16\%$ on the Thelonious phantom and $N = 20$ paths, the correction is below 2.5%.

Under any of these three conditions, $T_{\text{eff}} \rightarrow T_{\text{avg}}(\theta)$, and the local law reduces to $\text{APD}(\mathbf{r}) = \text{IPD} T_{\text{avg}}(\theta(\mathbf{r})) [\mu(\mathbf{r})]_+$.

D. Directional and polarization analysis on Thelonious

To complement the single-direction comparison reported in the main paper (Section V-C and Table IV), we evaluate the geometric law against the full polarization-aware Fresnel integration. The grid is 128 illumination directions sampled uniformly on the sphere and three polarization states: TE, TM, and unpolarized. The Thelonious phantom carries skin properties at 28 GHz with $\text{IPD} = 1 \text{ W/m}^2$. Figure S1 reports the results.

The TE and TM densities in panel (a) bracket the unpolarized density symmetrically, illustrating the $\frac{1}{2}(T_s + T_p)$ averaging that underlies T_{avg} . Panel (b) shows that the total absorbed power clusters around $T_0 A_\perp$ across directions, with the directional spread (CV) set by self-shadowing rather than by polarization. Across the 128 directions, the unpolarized full-Fresnel total absorbed power matches the simplified prediction $T_0 A_\perp(\hat{\mathbf{k}})$ with mean bias -1.2% and standard deviation 0.5%. The largest single-direction deviation is 2.2%.

E. Grazing incidence

The pointwise comparison in the main paper (Table IV) excludes triangles with $\theta > 75^\circ$. The headline check uses a single plane wave incident from above, the most grazing-heavy incidence for an upright body. The area-weighted fraction of the illuminated surface at $\theta \geq 75^\circ$ is 50% from above, against 19% under frontal illumination and 26% on a convex sphere of equal area. These grazing triangles carry only about 15% of the absorbed power, because the cosine weight $[\mu]_+ = \cos \theta$ suppresses their contribution.

The local relative error grows beyond 75° , but this is an artifact of the vanishing APD: as $\cos \theta \rightarrow 0$ both the predicted and the reference value go to zero, so a small absolute difference becomes a large ratio. The absolute error at grazing stays at or below the near-normal band, at most about 2% of the peak APD. For this top-down incidence the geometric law matches the area-integrated total absorbed power to within about 0.35%.

S2. PSEUDO-BREWSTER COMPENSATION ACROSS TISSUES

A. Azzam's high-index criterion

Azzam [2] showed that for a lossless dielectric substrate with refractive index $|\tilde{n}| > 2 + \sqrt{3} \approx 3.73$, the unpolarized reflectance varies by less than 1% over $[0^\circ, 60^\circ]$. The criterion follows from a series expansion of $|r_s|^2 + |r_p|^2$ around the pseudo-Brewster angle $\theta_{pB} = \arctan |\tilde{n}|$. The leading angular dependence cancels at $|\tilde{n}|^2 = 2 + \sqrt{3}$, leaving a quartic residual.

TABLE S1: Pseudo-Brewster compensation on the three outer-body tissues at 28 GHz, IT'IS Cole–Cole values. The variation column gives the maximum deviation of T_{avg}/T_0 from unity over $[0^\circ, 60^\circ]$ and over $[0^\circ, 75^\circ]$.

Tissue	ϵ_r	σ [S/m]	$ \tilde{n} $	T_0	θ_{pB}	var.
Skin	16.55	25.82	4.84	0.536	78°	3.3%/5.7%
Fat	6.09	5.04	2.63	0.784	69°	4.2%/17%
Eye (vitreous humor)	28.81	50.69	6.59	0.426	81°	7.4%/20%

TABLE S2: Frequency-resolved Cole–Cole values for skin (IT'IS v5.0) showing T_0 , $\bar{T}$, $R = T_0/\bar{T}$, and the skin depth.

f [GHz]	ϵ_r	σ [S/m]	$ \tilde{n} $	T_0	$R = T_0/\bar{T}$
0.3	60.0	0.40	7.93	0.381	0.938
0.9	41.4	0.87	6.70	0.445	0.957
2.4	38.0	1.46	6.29	0.470	0.964
6.0	35.6	4.00	6.07	0.481	0.968
10	31.3	8.01	5.87	0.489	0.971
28	16.6	25.8	4.84	0.536	0.988
40	11.6	30.0	4.30	0.570	1.000
60	7.98	36.4	3.68	0.622	1.015
100	6.35	50.0	3.01	0.701	1.035

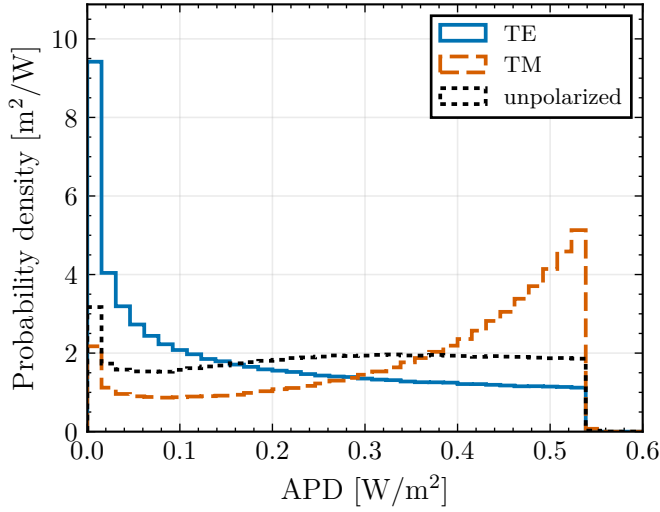
The criterion weakens for $|\tilde{n}| > 2.5$, where the unpolarized reflectance varies by less than 6% over $[0^\circ, 60^\circ]$. We extend the criterion empirically to lossy biological tissue $\tilde{n} = n - i\kappa$ and verify on the IT'IS catalog.

B. Quantitative evaluation across outer-body tissues

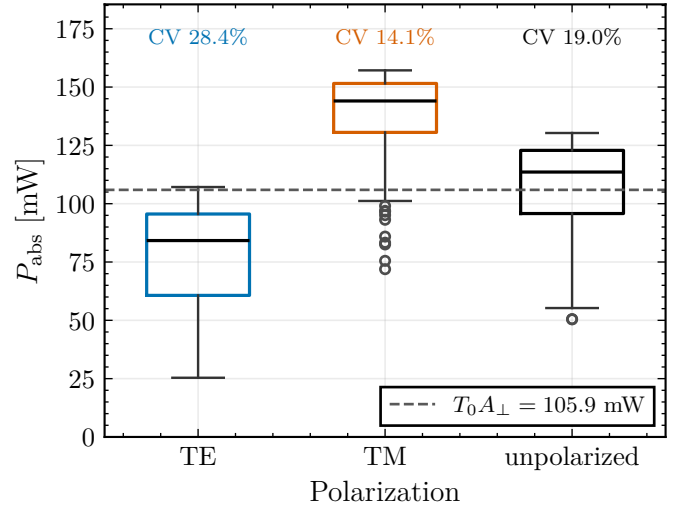
At body surfaces relevant to regulatory dosimetry above 6 GHz, only three tissues can be the outermost layer: skin (the dominant case), subcutaneous fat (rare, exposed at the abdomen and thighs), and the eye. Table S1 reports $\tilde{n}$, T_0 , the pseudo-Brewster angle, and the maximum angular variation of T_{avg}/T_0 on each of these three tissues at 28 GHz, with the IT'IS v5.0 Cole–Cole catalog as the source for ϵ_r and σ [3], [4]. Over $[0^\circ, 60^\circ]$ the variation is below 5% on skin and fat and below 8% on vitreous humor. Beyond 60° the high-index tissues (skin, vitreous) bend upward toward their pseudo-Brewster transmission peak, while fat ($|\tilde{n}| = 2.63$, near the Azzam threshold) drops monotonically. The cosine weight $\mu(\mathbf{r})$ in the surface integral suppresses contributions from $\theta > 60^\circ$, so the table value at 60° is the operationally relevant bound. Figure S2 shows this directly.

C. Frequency window of validity

Across 0.3–100 GHz, the IT'IS Cole–Cole tail keeps skin $|\tilde{n}|$ above 3 (Table S2). The pseudo-Brewster crossover where $T_0/\bar{T} = 1$ is at 40.4 GHz. Below this frequency T_0 underestimates the flux-weighted absorbed power (conservative for compliance). Above, it overestimates by at most 3.5% at 100 GHz. The root-mean-square error stays below 5% across the band. Figure S3 shows the angular profile $T_{\text{avg}}(\theta) \cos \theta$ at six representative frequencies. The family is nearly indistinguishable from the simplified $T_0 \cos \theta$ across the wireless band.



(a) Pointwise on illuminated triangles.



(b) Across 128 directions.

Fig. S1: Directional and polarization analysis on the Thelonious phantom (skin at 28 GHz, IPD = 1 W/m²). (a) Probability density of local APD on illuminated triangles for TE, TM, and unpolarized illumination. The unpolarized density is the symmetric mean of the two polarized densities. (b) Distribution of total absorbed power P_{abs} across 128 directions. The dashed line marks the prediction $T_0 A_{\perp} = 105.9$ mW. The coefficient of variation across directions is annotated above each box.

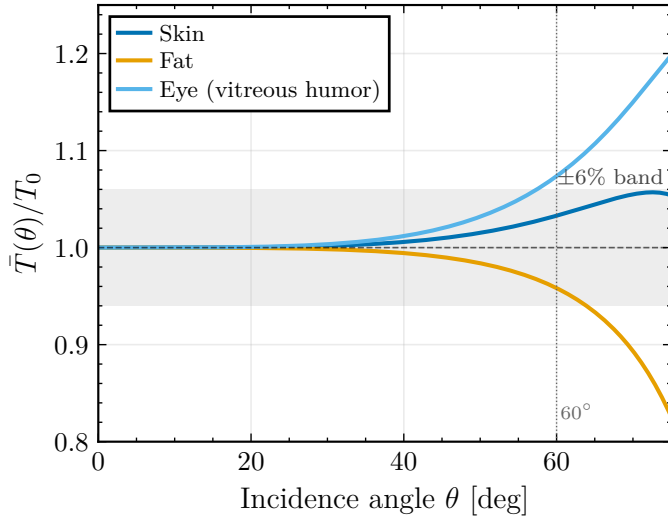


Fig. S2: Pseudo-Brewster compensation on the three tissues that can be the outermost layer at body surfaces of regulatory interest above 6 GHz: skin, subcutaneous fat, and ocular vitreous humor at 28 GHz, IT'IS Cole–Cole properties. The shaded band marks $\pm 6\%$. All three curves stay inside the band over $[0^\circ, 60^\circ]$. Beyond 60° , vitreous humor (high index, pseudo-Brewster near grazing) bends upward and fat (near the Azzam threshold) drops; skin stays within $\pm 6\%$ across the entire $[0^\circ, 75^\circ]$ range.

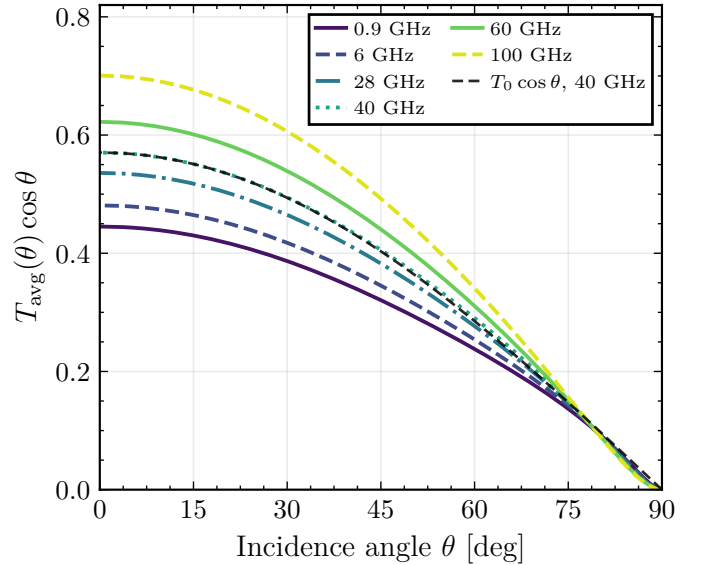


Fig. S3: Pseudo-Brewster compensation across the wireless band on skin: $T_{\text{avg}}(\theta) \cos \theta$ at six frequencies from 0.9 to 100 GHz, with the simplified $T_0 \cos \theta$ prediction at 40 GHz overlaid (dashed). The collapse to a near-universal geometric profile is the visual content of the pseudo-Brewster compensation across the band.

S3. LAYERED TISSUE MODEL AND THE CUBE SAR

A. Frequency transition

Three dimensionless parameters govern the transition between the thin-skin and the layered cube SAR regimes. The depth parameter $2\alpha L$ compares the SAR penetration

depth to the cube side $L = 21.5$ mm. The depth efficiency $h(2\alpha L)$ is the fraction of the surface SAR surviving mass-averaging to depth L . The capture fraction $C = 1 - e^{-2\alpha L}$ is the fraction of transmitted power absorbed within the cube depth. At 28 GHz, $2\alpha L \approx 47$ for skin and $C \approx 1$: the thin-skin shortcut applies. At 2.45 GHz, $2\alpha L \approx 1.9$ and $C \approx 0.85$, with 15% of transmitted power passing

TABLE S3: Tissue properties at 2.45 GHz from the IT'IS catalog [3], [4].

Tissue	ϵ_r	σ [S/m]	n	κ	δ_{SAR} [mm]
Skin	38.0	1.46	6.22	0.860	11.3
Fat	5.28	0.105	2.30	0.166	58.5
Muscle	52.7	1.74	7.31	0.873	11.2

through the cube into deeper tissue. Below 1 GHz, $2\alpha L$ falls to order unity and the homogeneous half-space breaks down because the field passes through multiple tissue layers within the cube.

B. Three-layer model and Chew recursion

A three-layer planar model is used: skin (epidermis + dermis, $d_1 \approx 2$ mm), subcutaneous fat ($d_2 \approx 5$ –20 mm), semi-infinite muscle. At 2.45 GHz the three layers carry contrasting dielectric properties:

Fat is much less lossy than skin and muscle ($\delta_{\text{SAR}} = 58.5$ mm). The wave passes through a thick fat layer with little attenuation and reaches the fat-muscle interface, where the impedance mismatch produces a strong backward wave with reflection $|r_{2,3}| \approx 0.52$.

Define the normal-incidence Fresnel coefficient at the interface between layers j and $j+1$,

$$r_{j,j+1} = \frac{\tilde{n}_j - \tilde{n}_{j+1}}{\tilde{n}_j + \tilde{n}_{j+1}}, \quad (\text{S10})$$

and the one-way propagation factor across layer j ,

$$P_j = e^{-ik_0 \tilde{n}_j d_j} = e^{-\alpha_j d_j} e^{-i\beta_j d_j}, \quad (\text{S11})$$

with $\alpha_j = k_0 \kappa_j$ and $\beta_j = k_0 n_j$. The generalized reflection coefficient at each interface follows the Chew recursion [5],

$$\tilde{\Gamma}_3 = r_{2,3}, \quad (\text{S12})$$

$$\tilde{\Gamma}_2 = \frac{r_{1,2} + \tilde{\Gamma}_3 P_2^2}{1 + r_{1,2} \tilde{\Gamma}_3 P_2^2}, \quad (\text{S13})$$

$$\tilde{\Gamma}_1 = \frac{r_{0,1} + \tilde{\Gamma}_2 P_1^2}{1 + r_{0,1} \tilde{\Gamma}_2 P_1^2}, \quad (\text{S14})$$

with $r_{0,1} = (1 - \tilde{n}_1)/(1 + \tilde{n}_1)$. The overall power transmission coefficient replacing T_0 in the cube SAR is

$$T_{\text{lay}} = 1 - |\tilde{\Gamma}_1|^2. \quad (\text{S15})$$

Figure S4 shows $T_{\text{lay}}(f)$ across 0.3–10 GHz for the canonical 2 mm skin / 10 mm fat / muscle stack against the skin-only baseline $T_0(f)$. The layered transmission departs strongly from the half-space prediction in this regime. A fat-transparent peak near 1.3 GHz appears where the 10 mm fat layer is electrically thin and the air-skin interface is the only significant reflector. A deep dip near 3.3 GHz appears where the fat layer is approximately a quarter-wave thick and the round-trip phase through fat flips the impedance seen at the air-skin interface back into a strong mismatch. The half-space T_0 curve, by construction, reproduces none of this structure. The departure motivates the layer-resolved Chew formulation throughout this section.

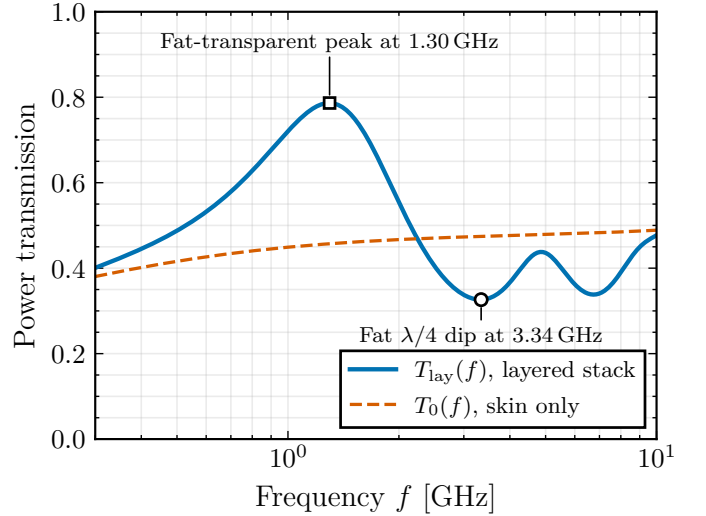


Fig. S4: Layered power transmission $T_{\text{lay}}(f)$ at normal incidence for the 2 mm skin / 10 mm fat / semi-infinite muscle stack, IT'IS v5.0 Cole–Cole properties, with the skin-only $T_0(f)$ baseline overlaid. The fat layer's electrical thickness sweeps through quarter-wave and half-wave conditions across the band, producing a transparent peak near 1.3 GHz and a $\lambda/4$ dip near 3.3 GHz that the homogeneous half-space model misses entirely.

C. Standing-wave SAR

Inside layer j , with z measured from the top of that layer, the electric field has the form

$$E_j(z) = A_j [e^{-ik_j z} + g_j e^{+ik_j z}], \quad k_j = k_0 \tilde{n}_j, \quad (\text{S16})$$

with $g_j = \tilde{\Gamma}_{j+1} P_j^2$ the backward-to-forward amplitude ratio at the top of the layer ($g_3 = 0$ for the semi-infinite muscle). The amplitudes are linked by continuity of the tangential field at each interface,

$$A_{j+1}(1 + g_{j+1}) = A_j e^{-ik_j d_j} [1 + g_j e^{2ik_j d_j}], \quad (\text{S17})$$

and A_1 follows from the incident amplitude through the air-skin transmission $1 - |\tilde{\Gamma}_1|^2$.

The squared modulus expands into three terms,

$$|e^{-ik_j z} + g_j e^{ik_j z}|^2 = e^{-2\alpha_j z} + |g_j|^2 e^{+2\alpha_j z} + 2|g_j| \cos(2\beta_j z + \psi_j), \quad (\text{S18})$$

with $\psi_j = \arg g_j$. The SAR in layer j is therefore

$$\text{SAR}_j(z) = \frac{\sigma_j |A_j|^2}{2\rho_j} [e^{-2\alpha_j z} + |g_j|^2 e^{+2\alpha_j z} + 2|g_j| \cos(2\beta_j z + \psi_j)]. \quad (\text{S19})$$

The three terms are forward decay, backward growth, and the coherent interference between them. The half-space limit recovers when $g_j = 0$.

Figure S5 renders (S19) for the canonical 2 mm skin / 10 mm fat / muscle stack at 2.45 GHz under unit-intensity normal incidence. The SAR stays high in the lossy skin layer because the high conductivity absorbs the incident

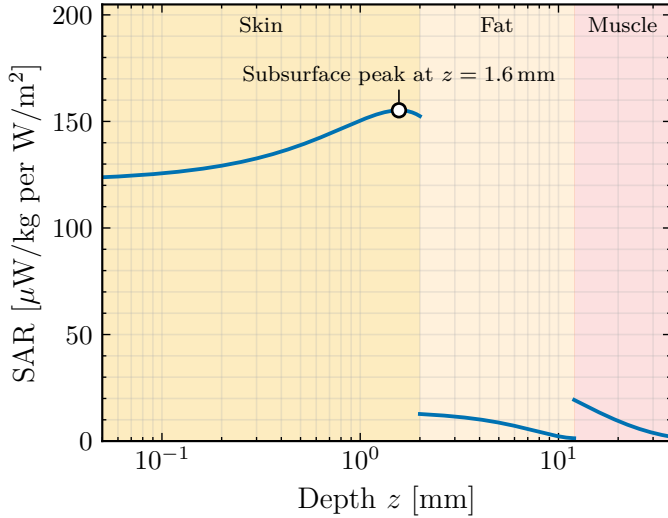


Fig. S5: Standing-wave SAR profile $SAR(z)$ across 2 mm skin / 10 mm fat / semi-infinite muscle at 2.45 GHz, normal incidence, IT'IS v5.0 properties. The shaded bands mark the three layers. The subsurface peak (open circle) is set by the standing-wave structure of (S19). For this stack it falls inside the skin layer.

flux quickly. It then drops several orders of magnitude in the low-loss fat, then rises again as the wave couples into muscle. The subsurface peak at $z \approx 1.6$ mm is inside skin, in agreement with the criterion of Eq. (S27) below.

D. Cube integral

The integral of $SAR_j \rho_j$ over the full thickness of a finite layer j admits a closed form,

$$\int_0^{d_j} SAR_j \rho_j dz = \frac{\sigma_j |A_j|^2}{2} \left[\frac{1 - e^{-2\alpha_j d_j}}{2\alpha_j} + |g_j|^2 \frac{e^{2\alpha_j d_j} - 1}{2\alpha_j} + \frac{|g_j|}{\beta_j} (\sin(2\beta_j d_j + \psi_j) - \sin \psi_j) \right]. \quad (S20)$$

For the semi-infinite muscle ($g_3 = 0$, depth $d_3 = L - d_1 - d_2$) only the forward term survives,

$$\int_0^{d_3} SAR_3 \rho_3 dz = \frac{\sigma_3 |A_3|^2}{2} \frac{1 - e^{-2\alpha_3 d_3}}{2\alpha_3}. \quad (S21)$$

The peak spatial-average SAR over the cube is the sum of the layer contributions divided by the total tissue mass,

$$\langle SAR \rangle_{\text{cube}} = \frac{1}{m} \sum_{j=1}^3 \int_0^{\min(d_j, z_j^{\max})} SAR_j \rho_j dz. \quad (S22)$$

E. APD bound on the cube SAR

The integral form (S22) combined with the cube placement search returns the exact psSAR_{10g} in the planar three-layer model. For compliance certification, an energy-conservation argument gives a closed-form upper bound that depends only on the local APD, with no Chew recursion or volumetric search.

a) *Statement.*: Let $L = (m/\rho_m)^{1/3} = 21.5$ mm denote the IEC/IEEE 62704-1 cube side at $m = 10$ g and $\rho_m = 1000$ kg/m³, and let APD denote the absorbed power density at the cube's body-surface footprint. Then

$$\text{psSAR}_{10g} \leq \frac{g \text{APD}}{\rho_m L}, \quad 1 \leq g \leq \sqrt{2}, \quad (S23)$$

where g is a geometric prefactor set by the angle between the cube axes and the local body normal $\hat{n}$. The tightest case $g = 1$ holds when the cube z-axis aligns with $\hat{n}$, the natural alignment in the planar three-layer model. The worst case $g = \sqrt{2}$ is the geometric maximum cube cross-section by a plane.

b) *Derivation.*: Per unit body-surface area, the time-averaged flux entering tissue is APD by definition. The cube intercepts this flux over its body-surface footprint A . Total absorbed power inside the cube cannot exceed total flux entering through A , by energy conservation:

$$P_{\text{abs,cube}} \leq \text{APD} \cdot A. \quad (S24)$$

Dividing by cube mass $m = \rho_m L^3$ and substituting $A = gL^2$ returns (S23). Equality in (S24) requires every joule entering the footprint to deposit inside the cube, which holds in the strong-attenuation limit $\alpha L \rightarrow \infty$.

c) *Body-cube intersection geometry.*: The footprint $A = gL^2$ depends on the cube orientation relative to the body normal. For an axis-aligned cube whose z-axis aligns with $\hat{n}$, the body plane cuts perpendicular to a cube face and the intersection is the constant $L \times L$ square: $g = 1$. The geometric maximum cross-section of an axis-aligned cube by an arbitrary plane is achieved by a face-diagonal cut (plane normal $(1, 1, 0)/\sqrt{2}$) and equals $\sqrt{2} L^2$, giving $g = \sqrt{2}$. The body-diagonal hexagonal cut (plane normal $(1, 1, 1)/\sqrt{3}$) is smaller at $3\sqrt{3}/4 \approx 1.30$, so the face-diagonal rectangle sets the universal ceiling.

The IEC/IEEE 62704-1 mass constraint tightens this. The $\sqrt{2} L^2$ cross-section occurs at the cube center, where each half holds 50% of the cube volume; that placement violates the 90%-tissue rule. Pushing the cube deeper into tissue to recover the 90% rule moves the body cut from the cube center toward an edge, where the cube cross-section is smaller. The realistic worst-case g under the IEC mass rule sits between 1 and $\sqrt{2}$, depending on the tilt between cube axes and $\hat{n}$.

d) *Tightness.*: For a homogeneous absorbing half-space with field amplitude decay rate α , the SAR depth profile is $SAR(z) = SAR(0) e^{-2\alpha|z|}$ and energy conservation pegs the surface SAR to the absorbed flux, $\rho_m SAR(0) = 2\alpha \text{APD}$. The face-flush cube average evaluates to

$$\langle SAR \rangle_{\text{cube}} = \frac{\text{APD}}{\rho_m L} (1 - e^{-2\alpha L}), \quad (S25)$$

so the bound (S23) at $g = 1$ is loose by the capture-fraction factor $C(\alpha L) = 1 - e^{-2\alpha L}$. Table S4 lists C on skin across frequency. Above 5 GHz the bound is exact to within a few percent. Below 1 GHz the bound is loose by approximately 50% on skin and more on fat where the field penetration depth exceeds the cube side.

TABLE S4: Capture fraction $C(\alpha L) = 1 - e^{-2\alpha L}$ on skin, where α is the field amplitude decay rate and $L = 21.5$ mm.

f [GHz]	$1/(2\alpha)$ [mm]	$2\alpha L$	C
0.9	19.9	1.1	0.67
2.45	11.3	1.9	0.85
5	5.3	4.1	0.98
10	1.9	11.3	1.00
28	0.46	46.7	1.00
60	0.24	89.9	1.00

A Monte Carlo cube search over 15 surface points on the Thelonious mesh, four polarization configurations, and five incidence directions per point across 1–28 GHz returns a median bound-to-numerical ratio of 1.30 and a maximum of 1.46, consistent with the combined effect of the capture fraction and the geometric prefactor g .

e) Numerical evaluation at ICNIRP limits.: With $\rho_m L = 21.5$ kg/m² and the worst-case $g = \sqrt{2}$, the bound at the ICNIRP basic restriction $APD \leq 10$ W/m² evaluates to $\text{psSAR}_{10g} \leq 0.66$ W/kg. The ICNIRP basic restrictions on psSAR_{10g} are 2 W/kg for head and trunk and 4 W/kg for limbs (general public, 6-min average [1]). The bound is below these limits by factors of 3.0 and 6.1 respectively. With the planar-aligned case $g = 1$, the bound tightens to 0.47 W/kg, with margins of 4.3 and 8.6.

f) Implications.: The bound (S23) reduces psSAR_{10g} compliance certification to APD compliance. Anywhere the closed-form chain delivers $APD \leq 10$ W/m², the cube quantity sits below 0.66 W/kg automatically, with margin. The volumetric cube search and the layered field profile remain useful for verification or for extracting an explicit psSAR_{10g} value, but play no role in certification. ICNIRP's basic-restriction switch from psSAR_{10g} (sub-6 GHz) to APD (above 6 GHz) reflects the historical metric and measurement convention rather than a physical necessity: the surface quantity certifies both restrictions across 1–100 GHz.

F. Subsurface peak criterion

In the homogeneous half-space, the SAR maximum is at $z = 0$ because $e^{-2\alpha z}$ is monotonic. Differentiating (S19) in skin (layer 1) at the surface,

$$\left. \frac{d\text{SAR}_1}{dz} \right|_{z=0} = \frac{\sigma_1 |A_1|^2}{2\rho_1} [-2\alpha_1 + 2\alpha_1 |g_1|^2 - 4\beta_1 |g_1| \sin \psi_1]. \quad (\text{S26})$$

Dividing by $2\alpha_1 > 0$ gives the criterion for a subsurface SAR maximum,

$$|g_1|^2 - \frac{2\beta_1}{\alpha_1} |g_1| \sin \psi_1 > 1. \quad (\text{S27})$$

The criterion has two contributions. The reflection-strength term $|g_1|^2$ rarely exceeds unity on its own. Typical values at 2.45 GHz are $|g_1| \approx 0.6$, giving $|g_1|^2 \approx 0.37$. The interference term $-(2\beta_1/\alpha_1)|g_1| \sin \psi_1$ dominates because $\beta_1/\alpha_1 = n_1/\kappa_1 \approx 7$ for skin at 2.45 GHz, and even modest reflections $|g_1| \approx 0.6$ with negative ψ_1 produce a large positive contribution.

For a 2 mm skin / 10 mm fat / muscle stack at 2.45 GHz, $|g_1| = 0.60$, $\psi_1 = -60^\circ$, and the criterion evaluates to $0.36 + 7.5 = 7.9 \gg 1$. Thin lossy skin over thick low-loss fat over lossy muscle gives a Fabry–Pérot enhancement of order $[(1 + |g_1|)/(1 - |g_1|)]^2$ at standing-wave antinodes, factors of 2–4 for typical anatomy.

G. Validity limits of the layered model

Three phenomena invalidate the layered planar half-space at sufficiently low frequencies.

a) Whole-body resonance, below approx. 300 MHz.:

When the free-space wavelength becomes comparable to body height ($\lambda/2 \approx 0.85$ m at approximately 175 MHz, with the resonance peaked near 70 MHz for a grounded standing adult [6]), the human body acts as a resonant dipole antenna. Internal fields are dominated by whole-body current distributions, not by the surface-transmitted plane wave. Localized hotspots form at anatomical constrictions (ankles, wrists, neck) where current density is highest. These hotspot locations bear no relation to the surface APD. The surface-to-volume reduction does not apply.

b) Finite-thickness anatomy.: Thin features such as the ear pinna (cartilage thickness approximately 2 mm) violate the semi-infinite half-space assumption even at higher frequencies. At 2 GHz, $\alpha d \approx 0.09$ for a 2 mm ear, so the one-way attenuation is only 0.91. The transmitted field partially exits the far side and illuminates the opposite surface. The SAR in such structures requires a finite-slab transmission model rather than a half-space model.

c) Body-traversing multipath.: At frequencies where δ_{SAR} exceeds the body diameter at narrow cross-sections (e.g., $\delta_{\text{SAR}} \approx 20$ mm at 900 MHz versus forearm diameter approximately 60 mm), the field transmitted through one side reaches the opposite side. Reflections from the far body-air interface create a second backward wave, and the resulting interference pattern can produce deep hotspots at locations not predictable from the surface fields alone.

S4. MIE RESIDUAL ANALYSIS

For a lossy sphere of radius a and complex refractive index $\tilde{n}$, the Mie series gives an exact absorption efficiency $Q_{\text{abs}}(x, \tilde{n})$ where $x = \pi d/\lambda$ is the size parameter. The geometric law predicts $Q^{\text{geom}} = T_0$, and the relative error is $(T_0/Q_{\text{abs}} - 1) \times 100\%$.

The error decomposes as

$$\frac{T_0}{Q_{\text{abs}}} - 1 = \underbrace{\left(T_0 / \langle T_{\text{avg}} \rangle_{\text{sphere}} - 1 \right)}_{\text{Fresnel}} + \underbrace{\left(\langle T_{\text{avg}} \rangle_{\text{sphere}} / Q_{\text{abs}} - 1 \right)}_{\text{diffraction}}, \quad (\text{S28})$$

with the Fresnel term shape- and frequency-dependent but size-independent (the sphere ratio $R(f) = T_0/\langle T_{\text{avg}} \rangle$ that crosses unity at approximately 39 GHz on skin), and the diffraction term scaling as $x^{-2/3}$ in the optical regime, with the wave bending into the geometric shadow and adding absorption that the surface law misses.

TABLE S5: Mie residual $T_0/Q_{\text{abs}} - 1$ in percent on lossy spheres with frequency-dependent IT'IS skin properties. Negative values mean the surface law underestimates absorption. The 1 GHz finger entry is in the Rayleigh regime ($x = \pi d/\lambda = 0.18$), where the surface law is invalid and the residual is large and positive. It is included only as a regime marker.

f [GHz]	Finger (17 mm)	Arm (80 mm)	Head (180 mm)	Torso (300 mm)
1	+1260 (Rayleigh)	-60	-53	-44
3	-68	-47	-35	-29
6	-61	-37	-27	-21
10	-53	-30	-22	-17
28	-38	-20	-14	-10
60	-27	-11	-6.1	-3.8
100	-19	-5.1	-1.2	+0.4

TABLE S6: Integrated diffraction effect on whole-body absorbed power on the Thelonious phantom: $(P_{\text{diff}} - P_{\text{simp}})/P_{\text{simp}}$ in percent, at $S_{\text{inc}} = 1 \text{ W/m}^2$ on skin.

f [GHz]	$(P_{\text{diff}} - P_{\text{simp}})/P_{\text{simp}}$
1	+33%
3	+12%
6	+6.0%
10	+3.5%
15	+2.3%
28	+1.2%
40	+0.8%
60	+0.6%
77	+0.4%
100	+0.3%

Table S5 reports the residual at seven frequencies on four representative body-part diameters with frequency-dependent IT'IS skin properties. For body-scale objects (head, torso), the framework error is 10–14% at 28 GHz and 4–6% at 60 GHz, dominated by diffraction into the geometric shadow rather than by the Fresnel approximation. At 100 GHz the head and torso residuals are sub-percent. Fingers are the worst case, with errors above 20% across the entire mmWave band. The sphere is the worst-case geometry for a surface law: its uniform curvature maximizes both Fresnel and diffraction errors. The human body has large flat regions on the torso, and net errors stay smaller. Fig. 8 of the main paper shows the empirical match against volunteer and FDTD data.

A. Integrated diffraction effect on Thelonious

Table S6 reports the diffraction effect on whole-body absorbed power across frequency on the Thelonious phantom. We compute P_{diff} by replacing $[\mu]_+$ with the GELU smoothing in Eq. 16 of the main paper and adding the second-order curvature term $T_0(H/k) \text{GELU}(\mu)^2$. The baseline $P_{\text{simp}} = \text{IPD } T_0 A_{\perp}(\hat{\mathbf{k}})$ uses the same T_0 , so the entry isolates diffraction. Six cardinal directions are averaged. Mean curvature H is taken from the cotangent Laplacian on the merged-vertex mesh.

The effect falls below 1% above 30 GHz and below 0.5% above 60 GHz on body-scale geometry. Below 6 GHz it

TABLE S7: Worst-case compliance threshold across body sizes at 28 GHz with $\bar{T} = 0.543$.

Body type	m [kg]	h [cm]	A [m ²]	m/A [kg/m ²]	IPD_{max} [W/m ²]
Infant (1 yr)	8	70	0.379	21.1	6.2
Child (6 yr)	20	110	0.776	25.8	7.6
Adolescent	50	160	1.491	33.5	9.9
Adult (ref.)	70	170	1.810	38.7	11.4
Large adult	100	180	2.193	45.6	13.5

grows quickly. At 1 GHz the body diameter is comparable to the wavelength and the geometric law breaks down, consistent with the Mie regime identified in the main paper.

S5. ANTHROPOMETRIC SCALING

All cross-references in this section that take a ‘main paper’ qualifier point at the main paper’s labels. The whole-body compliance threshold there, evaluated at the worst-case directional exposure (Remark ??), is

$$\text{IPD}_{\text{max}} = \frac{0.16 m}{\bar{T} A}, \quad (\text{S29})$$

where A is the body surface area. The convex-hull bound on the absorption directivity cancels the ambient-occlusion factor $\bar{\eta} = A_{\text{ab}}/A$ in the worst-case algebra, leaving A rather than A_{ab} in the denominator. This is the conservative form used for regulatory bounds.

A. Du Bois scaling of body surface area

The empirical Du Bois formula [7] gives

$$A = 0.007184 m^{0.425} h^{0.725}, \quad (\text{S30})$$

in m² with mass m in kg and height h in cm. The threshold scales as m/A , so

$$\text{IPD}_{\text{max}} \propto \frac{m}{A} \propto m^{0.575} h^{-0.725} = \text{BMI}^{0.575} h^{0.425}, \quad (\text{S31})$$

recovering the dosimetry-literature observation [8], [9] that absorption cross-section scales with surface area while mass scales with volume.

B. Population evaluation

Table S7 evaluates (S29) on a representative population at 28 GHz with $\bar{T} = 0.543$. The threshold varies by a factor of approximately two between an infant and a large adult.

The reference adult clears the ICNIRP general-public reference level of 10 W/m² at 28 GHz under worst-case directional exposure, while smaller body sizes carry lower thresholds.

C. Polarization worst case

Linearly polarized illumination at the most sensitive direction shifts $\bar{T}$ in (S29) by at most a fraction $|\hat{B}|/(2\hat{A}_{\perp}) \leq D_B$, with $D_B = 16\%$ on the Thelonious anatomical phantom and $D_B \leq 27.9\%$ on the worst-case convex shape (an infinite cylinder at broadside). For multipath environments with $N \geq 20$ independent path orientations, the correction

is below 2.5% via the $D_B/\sqrt{2N}$ scaling of Section S1. The unpolarized (S29) is therefore conservative for any reverberation chamber and for the most common exposure scenario, a vertically polarized base-station illumination of a standing person.

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